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Weaponization of Trade: China's Rare Earth Monopoly and its Impact on US Defence Procurement

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ABSTRACT

This paper examines the intersection of geopolitics and trade in the rare earth elements industry with a focus on the United States, China and Australia. Imperative to defence technologies, REEs are a group of 17 elements possessing unique magnetic and luminescent properties that have emerged as a tool for economic leverage in the 21st century . The research investigates the interplay between market

conditions, and state policies that have led to a decline in the United States and a subsequent rise in China. The study further investigates key policies, particularly the import substitution strategy, to counter China's dominance.

Adopting a mixed-method approach, the study combines econometric analysis with qualitative insights from institutional reports, etc. The results suggest that geopolitical shocks are associated with short term fluctuations in trade flows. However, the absence of a long run phenomenon suggests that these disruptions do not translate into structural shifts in global supply chains. Within this framework, Australia's role appears more limited than what the policy suggests. The study suggests that Australia's role thus far, can only be understood as that of an alternative supplier rather than an emerging power that can dominate the market.

By situating REEs at the intersection of strategy and economy, this study explores how, more than resource abundance, efficiency governs global trade.

KEYWORDS- Rare Earth elements, defence technologies, monopoly, global trade, strategic vulnerabilities, supply chains

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Chapter 1

1.1 Introduction

“The Middle East has oil, China has rare earth.”

- Deng Xiaoping, 1992

When the former Chinese leader made this remark in the late twentieth century, it was interpreted primarily as an observation on China’s resource endowments rather than as an expression of strategic intent; this perception shifted in 2010, when China temporarily halted REE exports to Japan during the Senkaku Islands dispute. This episode demonstrated that economic interdependence, often viewed as a stabilizing force in the era of globalization, could also generate vulnerabilities.

Possessing unique magnetic, luminescent and catalytic characteristics, REEs are as indispensable to modern warfare in the 21st century as railroads were to 19th-century economies. Unlike oil or other contested resources, the trade of REEs presents a paradox. In the globalised world where comparative resource endowments dictate trade flows, the geological abundance of these minerals requires economies to rethink policies, regulations and state intervention to tackle this specific issue. The paradox is more pronounced in China’s rise from a marginal producer to the dominant supplier today. A common economic logic that persisted globally suggests that it is the ownership of resources that is essential for growth. This is reflected in the oil wars that continue to be fought even today, where the US is the dominant actor. However, in the case of REEs, their strategic power doesn't simply come from geological scarcity, but rather the economic scarcity in terms of lack of processing, manufacturing capacity, creating an industrial chokehold that China has curated since the late 20th century.

As of 2024, China produces 60% of the world’s REE but processes nearly 90%, making it the lead exporter. ([CSIS, 2024](#)). China’s dominance in REEs is neither accidental nor purely market-driven. It is a blend of sustained state intervention and trade policies that transformed the industry. To understand how REEs became a source of geopolitical power, it is necessary to trace

China's REE policies over time. The following section examines China's policies from 1975 to 2023, highlighting policies that have shaped its growth trajectory.

1.2 China's REE policies 1975-2023

China began mining, production and processing of REEs in the 1950s, but it wasn't until the 1970s that it became a prominent supplier. In the beginning, China focused on building an infant mining industry. It became increasingly clear that domestic consumption won't be enough to sustain growth in the sector, so with a mere trade of 150 to Japan of REE chlorides, China began exporting in 1973 ([Mancheri et al., 2019](#)). In 1980, this figure had increased to about 2000 t. In 1985, the US became an importer of REE from China when its capacity to export had reached 10000 t.([Park et al., 2023](#)) Growth in the sector was further encouraged through tax rebates that allowed firms to continue expanding their business into processing and manufacturing. At the same time, mines were shut down in the US, Japan and France due to strict environmental laws that made licenses extremely expensive. ([Park et al. 2023](#) ,[Zhanheng 2011](#))

However, environmental protection was not a priority for China, which, driven by low production costs and cheap labour, enabled rapid growth in this sector, amounting to 14% in 1990.

From the 1990s, problems such as illegal mining, biodiversity loss and disorder due to lenient policies became a matter of concern. ([Park et al. 2023](#) , [Wubekke, 2013](#),) In response, China recognized REEs as strategic minerals and installed minor regulatory controls. Foreign investment in the sector required permission from the Rare Earth Office. Licenses for exploration were suspended and renewed selectively favouring state owned enterprises. Despite the new regulations, enforcement remained uneven.

From 1999 onwards, up until 2009, was the period of export quotas aimed at controlling production and exports. During this period, export quotas favoured firms that produced high value products, echoing China's commitment to establishing an integrated value chain, while primary products saw a decline in quotas. During this period, tax rebates were reduced and readjusted and instead, export taxes were introduced and raised over time. As a result of this,

exports declined in 2005, but domestic consumption of downstream products such as magnets and electronics grew rapidly. ([Chen, 2010](#))

Between 2010 and 2018, China tightened controls even further by strengthening environmental regulations. This coincided with the WTO dispute between Japan and China, where China was ultimately accused of violating GATT rules. Following the WTO ruling, China shifted from trade based instruments to regulatory rules. A major reform was the consolidation of the REE industry into six large rare earth groups, which were either state owned or state controlled. The consequences of the state-owned REE industry were that in the periods that followed, China had greater power over its resources in terms of the information that was made available. The new era of REE statecraft involves instilling uncertainty by withholding information. This particular move has pushed the rest of the world to delve into alternative sources of REE supply.

Thus, by expansion into the value chain of REEs and the unconventional statecraft to feed domestic consumption, China poses strategic threats to economies, especially to non allied nations like the U.S., which are dependent on imports for sustaining vital industries like defence for national security. ([Mancheri et. al, 2019](#))

Chapter 2

2.1. Literature Review

2.1.1 RARE EARTH ELEMENTS AS STRATEGIC AND SECURITY CRITICAL RESOURCES

A number of scholarly works are dedicated to understanding the importance of REEs not only for their commercial uses but also for their applications in the defence sector. This is due to their wide-ranging applicability in the manufacture of sensors, radars, and other high-tech equipment. [Islam \(2025\)](#) talks about the significance of REEs in modern military systems, serving as the backbone of advanced technologies such as laser and precision guidance systems. ([Porcher et al., 2025](#)) Military-grade equipment especially requires exorbitant amounts of REEs, as referenced by [Chapman \(2017\)](#), who states that the F-35 Joint Strike Fighter uses 418 kg, and Virginia-class submarines use up to 4600 kg of REEs.

From 2000 to 2022, the world consumption of REEs products increased by 3.7 times, from 78 to 286 kt/year, with a compound annual growth rate of 6.1% [\(Peng, 2024\)](#). Despite the name “Rare Earth,” these elements are abundant in the Earth’s crust but are nevertheless classified as “critical” minerals. About 34 countries have been identified worldwide that are abundant in REEs; however, China retains its dominance in this sector, being the biggest exporter to the world by volume in 2019 [\(Mancheri et. al., 2019\)](#). Thus, the difficulty doesn't arise from the mining of REEs, which is done on a large scale globally; rather, the issue is the downstream processing and manufacturing of these minerals, where China holds a comparative advantage. [\(Vivoda, 2025\)](#), This downstream processing generates strategic dependence, rendering these minerals “critical” due to geopolitical vulnerability and supply risks.

The indispensable role of REEs has been widely echoed in scholarly literature. Research in the field has also persisted in discovering substitutes and recycling, but none have been fruitful so far [\(Hynek, 2023\)](#) [\(USGS, 2024\)](#). The unique magnetic, luminescent, and catalytic properties are

not easily replicated in alternative materials without losses in efficiency. Thus, they remain as the sole reliable source for modern weapons systems.

2.1.2 CHINA’S DOMINANCE IN THE RARE EARTH SUPPLY CHAINS

Throughout the 1970s and 80s, REE production was mainly concentrated in the Mountain Pass mine in California, prompted by government funding in research and development as a response to the arms race between the Soviet Union and the United States [\(Carrell, 2012,\).](#) As the industry expanded and corporations fought for patents, significant challenges emerged. In 1980, the IAEA enacted [“Part 40: Domestic licensing of source material.”](#) which reclassified some REEs as nuclear source materials, making mining more expensive due to licensing burdens. In the early 2000s, Mountain Pass was closed down due to corporations abandoning their production amidst increasing licensing and environmental costs. [This decline in US production occurred against the backdrop of a contrasting trajectory in China.](#)

The Bayan Obo (Inner Mongolia) mine was discovered in the 1920s, but was developed only after 1949(Park et al. 2023, Mancheri et al., 2019). China's near-monopoly over rare earth elements is widely understood in the scholarly literature as the outcome of a deliberate state-led developmental strategy rather than geological advantage or market forces alone(Naughton, 2007). In the late 20th century, China embarked on a phase of learning and capacity building(Wubbeke, 2013). While the US focused on innovative applications of REEs, China's research and development efforts focused on applied science.

The industrial base in China was underpinned by weak environmental controls, availability of cheap manual labour, lenient policies, and state supported infrastructure.(Park et al. 2023) Through subsidies, preferential loans, and local government support, Chinese firms consolidated dominance in processing and component production, creating structural dependence even when rare earth mining occurred outside China (Humphries, 2013). By the early 2000s, as global demand expanded, China had integrated into every stage of the REE value chain (Mancheri et al., 2019), making Chinese producers the only profitable ones in the market. Non Chinese

producers found it difficult to compete with China's strategic prices and subsequently exited the market, laying the groundwork for China's monopoly. (Htun, 2023). Zhao et al. (2016) discovered that Chinese rare earth firms possess a competitive advantage over their US counterparts in terms of cost structure and revenue profile. Furthermore, China's state-owned enterprises account for 76% of China's total investment in mining worldwide from 2005 to 2023(PwC, 2023).

Thus, the scholarly work suggests that China didn't merely outdo the rest of the world in REEs by being efficient but also by being patient and incurring huge fixed costs. The development of the REE industry came with major environmental losses that market-oriented economies of the West did not want to absorb. As a result, China's developmental strategy was building a comparative advantage through long term capacity accumulation, reinforcing China's dominant position through structural advantages.

2.1.3 RARE EARTH EXPORT CONTROLS AND STATECRAFT

China's REE export policies and statecraft have been a topic of contention throughout the 21st century. Introduced in 2003, China imposed export restrictions to restrict the quantity sold abroad and to feed domestic demand. These quotas were revised and tightened periodically to restrict foreign access while boosting the domestic industry. Hence, the export controls were installed to create "artificial" scarcity for geopolitical leverage. ([Wubekke, 2013](#)) This system was one of the core tools China used to shape global markets before 2015. China's reliance on REEs as statecraft became evident during the 2010 embargo to Japan over the Senkaku Islands dispute. ([Woods, 2025](#)) This was contested by the United States, European Union and Japan in 2012, and in 2014, the WTO released a report finding that China had indeed violated Article XI of the GATT and the Accession Obligation on limiting export restrictions that China had held while joining the WTO in 2001. On 1st January 2015, in response to the ruling, China removed its formal export quotas and duties for rare earths and replaced them with an export licensing system. The licensing system has also come under the radar, as it is deemed an indirect method of volume control. ([Hurst, 2010](#))

From 2020 onward, China shifted the governance of rare earth exports from ad hoc administrative measures to a formal "national security based" export control regime, anchored in the [Export Control Law of the People's Republic of China](#). This shift aimed to consolidate state authority over REEs legally, aligning trade with broader national security concerns. Thus, since 2021, China has stopped publishing annual export data, creating a lasting state of uncertainty: Foreign governments and companies lacking reliable information on Chinese output had to hedge against worse case scenarios, increasing their costs even without official supply reductions. ([Woods, 2025](#)) In October 2024, China consolidated control over the entire rare earth value chain, regulating mining, processing, magnet production, and related technologies under a unified national security framework. This culminated in 2025 with the securitization of exports through licensing restrictions on seven medium and heavy rare earths, including terbium and dysprosium,

which are critical for the defence sector.

China's near-monopoly status in REEs is best explained through standard economic concepts of control over the value chain rather than simple resource endowment alone. ([Mancheri et. al. 2019](#), [Hurst, 2010](#)) By vertically integrating into the downstream processing, China created high entry barriers even as global demand expands([Mancheri et al. 2019](#)). The way China sees it, its monopoly is not a result of the distortion of the market, but the market outcome of economies of scale and state-backed development.

Thus, the scholarly work as well as government reports suggest that the real power that China enjoys in the REE sector is not the export bans and other means of coercion, but in the logic that no competitor can enter the market at this stage without incurring huge losses that China was willing to incur decades earlier.

2.1.4 WEAPONIZED INTERDEPENDENCE

Influenced by the works of Newmann and Farrell, research in REEs places them in a broader framework of weaponized interdependence, which explains how states leverage asymmetric

positions in global economic markets to exert power. Their take on globalization suggests that network structures do not result in compatible cooperation nor in reducing the scale of asymmetry. Instead, power imbalances are heightened over time. Thus, they propose the idea of “asymmetric growth,” where globalization, like other networked forms of human activity, generates networks with stark inequality of influence. They argue that states occupying central positions in global networks—particularly those controlling chokepoints—can exploit these structures for strategic ends by restricting access or manipulating rules. Although their analysis focuses primarily on financial and technological networks, the framework is readily applicable to critical mineral supply chains, where concentration and lack of substitutes magnify coercive potential. (Shapiro et al., 2024; Newman and Farrell, 2019, USGS, 2024)

Building on this theoretical framework, [Woods \(2025\)](#) extends the logic of weaponized interdependence to REEs. He argues that export restrictions enacted by China shouldn't merely be seen as material constraints but as a form of informational statecraft wherein uncertainty is weaponized in the supply chain. [By withholding or classifying rare earth-related intellectual property, Beijing can create a polling equilibrium where outsiders cannot distinguish between a genuine breakthrough and strategic bluffing](#). This mechanism keeps uncertainty alive, and forces rivals to assume worst case scenarios leading to defensive overinvestment. [Woods\(2025\)](#) further discusses patent opacity and predicts that this will likely expand in the future as Beijing has incentives to safeguard and restrict its technological advancements to a greater degree than before.

[Kim and Rho's \(2024\)](#) analysis of the chip war shows how high technology sectors have become increasingly securitized due to supply chain interdependence. Though their work rests on analyzing semiconductors, the results can be applied to REEs as well. This study mirrors the United States' semiconductor import diversification toward other East Asian Tigers and examines a similar strategic shift in the REE supply chain, aimed at reducing reliance on a single dominant supplier.

Together, these works challenge former liberal ideologies of international trade that sought to reduce conflicts by increasing mutual costs. Instead, they suggest highly asymmetric market

structures in vulnerable sectors like defence can coerce without direct military confrontation. In the context of REEs, China's monopoly enables both material and symbolic forms of pressure on dependent states even in the absence of overt export bans. ([Blaire, 2023](#))

2.1.5 RARE EARTH STRATEGIC RIVALRY AND GREAT POWER COMPETITION

A central theme in REE literature is the great power competition between China and the US. Once treated as industrial inputs, they now occupy a central position in strategic rivalry between

the two nations. Even though the connection between economic prosperity and national security is not universally agreed upon, one cannot underscore the threats from China posing a vulnerability to the military sector (Hynek, 202; Vivoda, 2025). The literature investigates REEs not as commodities but as leverage embedded in the political economy of superpower competition.

China's high degree of control over REEs is perceived as a threat to US security, reflected in the Biden Administration's [100-day review report](#), which emphasizes the concentration of critical mineral supplies from a single nation, such as China, as a primary risk factor. (Brooks and Wohlforth, 2016) argue that China's rise to power doesn't simply rely on military expansion but rather on the newfound opportunity to dictate technological foundations in this sector. Perhaps, apart from the wide ranging direct military applications, REEs have also awarded China indirect means to control economies worldwide. Those who are dependent on REEs from China have echoed a shared sentiment of exploitation driven by China's monopoly, leading to a "new Cold War" which has the potential to bring down the US and other nations (Frum, 2012; Dobransky, 2011).

The strategic significance of REEs is heightened by signalling problems. Blair (2025) suggests that threats of export bans are as disruptive as actual supply shocks. This insight is built on the foundations of a weaponized interdependence form of trade, which operates on deterrence, uncertainty and anticipation rather than overt coercion. Based on this, Bridge (2008) makes the argument that resource power now not only depends on ownership of materials, but also control

over logistics and processing nodes. Brooks (2023) further notes that a contemporary superpower is shaped less by economic size and more by military preparedness. Klossek et al. (2016) investigated a variety of distortions in rare earth markets, including conflicting political and economic models, resource nationalism, market opacity, lack of trust, and inadequate collaboration, among others.

Together, the scholarly work delves into the dependence of the US on China for REEs, emphasizing how the cracks in the economic structure have exposed the defence sector to

vulnerabilities and undermined its ability to yield power on a global level. ([Vivoda, 2025](#))

2.1.6 US POLICY RESPONSES

Policy framework regarding REE goes as far back as the 1930s, when the US enacted the Strategic and Critical Minerals Stockpiling Act of 1939 ([SCMA](#)), which enables the United States government, through the Defence Logistics Agency, to purchase “critical minerals” (CMs) and reserve them in times of war. In the decades that followed, stockpiling targets continued to expand well after the end of World War II and into the Cold War, where the dominant policies at the time concentrated on military preparedness. By the late 20th century, changes in the global production patterns led to a reduction of stockpiled materials. ([Carrell, 2012](#),) The National Defence Stockpile was significantly reduced in the 1990s, even as China’s industrial base grew. [Butler \(2014\)](#) and [Hynek \(2023\)](#) emphasize that REEs became embedded in defense-critical systems precisely at the moment when U.S. production capacity declined, amplifying strategic vulnerability. From this perspective, stockpiling alone was increasingly viewed as insufficient, as modern defense systems depend on continuous access to refined materials rather than static reserves. The resurgence of REEs in the policy framework happened in the 2010s, driven by supply chain vulnerabilities. This later phase includes the periodic update of official “critical minerals” lists by the U.S. Geological Survey and other agencies, which have expanded dramatically. From a small set of Cold War-era materials to dozens of minerals, including REEs, lithium, and other high-tech metals, as part of federal strategy documents and executive directives

The Senkaku Islands dispute, as well as the 2019 bilateral trade dispute between the two nations, pushed the Department of Defence to reduce its dependence on Chinese exports. Most recently, the [John S. McCain National Defence Authorization Act for fiscal year 2019](#) prohibited DOD from procuring samarium cobalt and neodymium iron boron from “non allied nations”: Russia, China, Iran or North Korea. [Vivoda, Matthews, and Andresen \(2025\)](#) similarly note that rare earths are now embedded within a comprehensive critical minerals strategy, wherein state

intervention is justified not only by scarcity but by geopolitical risk and strategic competition.

[Islam \(2025\)](#) investigated the relationship between supply chain risks to defence expenditure, suggesting that these risks not only lead to increased spending or stockpiling but also domestic REE processing initiatives and partnerships with allied nations - further driving up military expenditure.

Reports by the U.S. Government Accountability Office (GAO) evaluate federal efforts to strengthen critical mineral supply chains and identify ongoing problems such as poor coordination between agencies, limited data, and the absence of long-term planning. These findings indicate that although critical minerals have received greater policy attention, implementation remains inconsistent and fragmented across government institutions. Efforts are being made to develop a domestic supply and revive the Mountain Pass mine, but to no avail since the mine resumed production in 2015 and was closed again that year due to competition from China. [Carrell\(2020\)](#) talks about the next best alternative, i.e., the Bear Lodge mine in Wyoming, which was developed in 2011 but was put on hold due to insufficient investment. Plenty of research is dedicated to finding substitutes and recycling methods, but none have produced robust results so far. [\(Hynek 2023\)](#)

[Shapiro et.al \(2024\)](#) discussed internal and external balancing strategies adopted by the InUS to counter China's dominance. Internally, the US has adopted Restrictive Investment screening mechanisms to limit Chinese access to critical mineral assets and technologies within its borders. Under CFIUS, foreign investments in Chinese state-owned enterprises are also subject to scrutiny. Externally, the US focuses on building partnerships with trusted allies to secure

alternative sources of critical minerals through cooperation with countries such as Australia, Canada and the United Kingdom, seeking to diversify supply away from China through joint trade investment. [\(Qing, 2023\)](#) By promoting cooperation through “like-minded” partners, the US aims to develop supply chains that are less vulnerable to supply shocks and geopolitical risks.

Overall, the literature shows that U.S. policy responses have shifted from legislative concerns to

a broader strategy that links defence planning, industrial policy, and supply chain resilience. However, scholars note that these efforts are still limited by complex regulations, uncertain market conditions, and continued reliance on foreign processing capacity.

2.1.7 IMPORT DIVERSIFICATION FROM CHINA TO AUSTRALIA

The study specifically focuses on the import diversification strategy adopted by the US to counter China's monopoly and to safeguard the defence sector from supply shocks. Although exports were diversified even during the 2000s, the need to move away from China as the dominant supplier only occurred after the 2010 WTO dispute. [Qing \(2023\)](#) analyzed 15-year data for China, Australia, India and Russia in terms of export competitiveness. The results indicated that Australia has demonstrated the highest growth rate in terms of the competitiveness of its rare earth industry, with a rate of 34.67%. China, India, and Russia follow closely behind with growth rates of 31.62%, 17.80% and 12.07%, respectively. Before 2010, Australia was less competitive than Russia, but it started to decline after surpassing Russia slightly in 2010. However, the difference in competitiveness between the two countries has been diminishing since 2017. It is evident that the international competitiveness of the rare earth industry in all four countries has shown a positive upward trend. Australia's competitiveness is improving in the industrial environment dimension, while its trade dimension is decreasing, but its infrastructure dimension consistently ranks second. At the same time, Russia and India rank third and fourth, respectively. ([Qing, 2023](#))

Thus, keeping in mind the strained relationship that the US and Russia have due to the Ukraine war and India's lack of a national framework for the REE value chain, Australia is the preferred

choice for the US for developmental deals in this sector. The allied states also share common goals that extend beyond mere trade deals. As of 2025, the US and Australia have signed an \$ 8.5 billion deal([White House, 2025](#)) to secure the REE supply, with plans to boost the industrial growth of REEs in Australia. Hence, the study will try to analyze whether Australia has the potential to surpass China as the dominant supplier for the US.

REEs have been a topic of great interest among scholars. As demonstrated by Xiao (2023), China dominates not just the global exports but also holds a central position in research work. With China's growing dominance culminating in a near monopoly, the research work has been dedicated to the subsequent risk to vital industries like defence. The literature on weaponized interdependence also suggests that it is important for the US to shift from China to allied nations till it revives its own industrial base. ([Qing, 2023](#)) The research tends to examine mining and trade flows without modelling a full multi-stage supply chain, where China's dominance is most prominent. Empirically, causal analysis of how China's policies affect global prices remains limited due to opacity. On the US side, strategies such as stockpiling, investments and public-private partnerships are mentioned but are rarely evaluated to understand their effectiveness. ([Hynek, 2023](#) , [Carrell, 2012](#).)

This study differs from existing literature by explicitly examining the vulnerability of the US defence supply chains to Chinese trade policies. It further incorporates Australia as an important diversification partner in trying to assess whether such strategies help in mitigating risks. In this sense, it aims to analyze whether the REEs deals between the US and Australia have been successful in countering China's monopoly.

Chapter 3

3. Methodology

This study employs a quantitative time series analysis to examine the dynamics between China and Australia in the REE export market for the United States. Given the high-tech applications of REE's in defence industries, understanding the nature of supplier relationships is critical. Thus, the Methodology is designed to capture both short term interactions and potential long-run relationships between the export shares of China and Australia for the US.

3.1. STATEMENT OF PROBLEM

Rare Earth Elements are indispensable to the defence sector. The United States of America relies heavily on China for imports of these “critical” minerals since the global supply is heavily concentrated in China, giving it a near-monopoly status. This dependence exposes the US to supply shocks and potential leverage from China. While the US has tried to mitigate these risks by import diversification away from China to allied nations like Australia, the effectiveness of this policy remains unclear. Understanding whether this strategy has proven successful is important for assessing national security, among other concerns.

3.2 SCOPE OF RESEARCH

The study focuses on REEs critical to the US, examining the US's decline in domestic production and subsequent rise in China's dominance and the import diversification strategy to Australia. The study focuses on the comparative analysis of industry decline and rise in the US and China from 1970 to 2000. Additionally, it uses import data to assess the effectiveness of import diversification to Australia from China from 1990 to 2024. Government reports and statements by the head of state are salient to the analysis of this study. Based on these findings, the study will try to analyze strategic risks posed to the defence sector.

3.3 AIMS AND OBJECTIVES

The aims and objectives of this study are:-

1. To analyze the degree of risks faced by the US defence sector from China's Monopoly
2. To evaluate the effectiveness of US efforts to diversify REE imports away from China to Australia.

3.4 RESEARCH QUESTION

The study attempts to answer the following research question: -

Has the strategy of import diversification to Australia in order to counter China's monopoly in REEs mitigated defence sector risks for the United States of America?

3.5 MATERIALS

The data sources used in this study are the United States International Trade Commission Dataweb (USITC). Data analysis will be carried out using quarterly import data from 1990 to 2024 for REEs, compounds and permanent magnets that are critical to the defence sector. For the scope of this study, the partner countries are Australia and China.

The analysis is based on the trade data classified under the following Harmonized Tariff schedule(HTS) codes-

- **2612**- Raw or processed ores specifically intended for extracting uranium or thorium.
- **2805**-Alkali or alkaline-earth metals; rare-earth metals, scandium and yttrium, whether or not intermixed or interalloyed; mercury
- **2846**- Rare-earth Metal Compounds, Inorganic Or Organic, Of Yttrium Or Scandium Or Of Mixtures Of These Metals
- **8505**- Electro-magnets; permanent magnets and articles intended to become permanent magnets after magnetization; electro-magnetic or permanent magnet chucks, clamps and similar holding devices; electro-magnetic couplings, clutches and brakes; electro-magnetic lifting heads; and parts thereof

The key variables of this study are:

- **CH_SHARE** - China's share of REE exports to the US

- **AUS_SHARE** - Australia's share of REE exports to the US
- **D_CH_SHARE**- First difference of China's export share to the US

The data is structured as a quarterly time series. Data on U.S. rare earth imports, disaggregated by partner country, are analyzed over a multi-year period to capture changes in sourcing patterns. The primary indicators include import quantities and partner-country shares, with particular attention to shifts in China's dominance and Australia's growing role as an alternative supplier.

The study also uses government reports, such as those from the US Geological Survey, to analyze global REE production and processing shares; Department of Defence reports to analyze strategic materials dependency; and Congressional Research Service reports to analyze policy changes, stockpiling, and legislative measures. These sources are necessary to understand the share of the defence sector consumption in total REE consumption by the US.

3.7 ECONOMETRIC METHODOLOGY

3.7.1 Stationarity Testing

The time series properties of the variables are examined using the Augmented Dickey Fuller test.

3.7.2 Autoregressive Model

To address autocorrelation, an AR(1) specification is introduced :

$$AUS_SHARE_t - AUS_SHARE_{t-1} = \alpha + \beta (AUS_SHARE_t - AUS_SHARE_{t-1}) + \rho (AUS_SHARE_t - AUS_SHARE_{t-1})_{t-1} + \epsilon_t$$

This allows the model to account for persistence in Australia's export share, reflecting the dynamic nature of supply adjustments.

3.7.3 ARDL Model

Given the mixed integration order of the variables, the ARDL model is employed.

The general equation of this model is :

$$\begin{aligned} \Delta \ln Y_{it} - \Delta \ln Y_{it-1} &= \alpha + \Delta \ln Y_{it-1} \\ \sum \beta_j \Delta \ln Y_{it-j} - \Delta \ln Y_{it-1} &+ \Delta \ln Y_{it-1} = 0 \\ \sum \gamma_j \Delta \ln Y_{it-j} - \Delta \ln Y_{it-1} &+ \epsilon_{it} \end{aligned}$$

The model is suitable as it captures both long run and short-run relationships and accommodates variables with mixed integrations.

Further, to test for the existence of a long run relationship, the ARDL bounds testing approach is applied with the null hypothesis of a no long run relationship, which will be decided upon by comparing the F statistic with critical bounds.

3.8 CONTROLS

To isolate the effects of diversification from broader structural trends, the analysis incorporates the following controls:

1. Major policy shocks, including China's export quota regime, WTO rulings, and 2020–2025 export controls
2. Consumption of REE for uses other than defence, like electronics, etc.
3. Price volatility periods, which may affect sourcing decisions independently of strategy

3.9 REPLICABILITY AND TRANSFERABILITY

- All HS/SITC codes used in the analysis are explicitly documented
- Data extraction procedures from USITC DataWeb are standardized and reproduced.

3.10 LIMITATIONS

The key limitation of this study is the unavailability of data that explicitly ties in imports for the defence sector by end use applications, distinguishing between defence and civilian uses.

The study is limited by diagnostic issues common in quarterly trade time-series data, including autocorrelation, heteroskedasticity, and some correlation among lagged

variables due to persistence in trade flows. To address dynamic adjustments, AR(1) and ARDL models were used.

Chapter 4

4.1 Stationarity Analysis

4.1.1 AUGMENTED DICKEY FULLER TEST FOR CH_SHARE

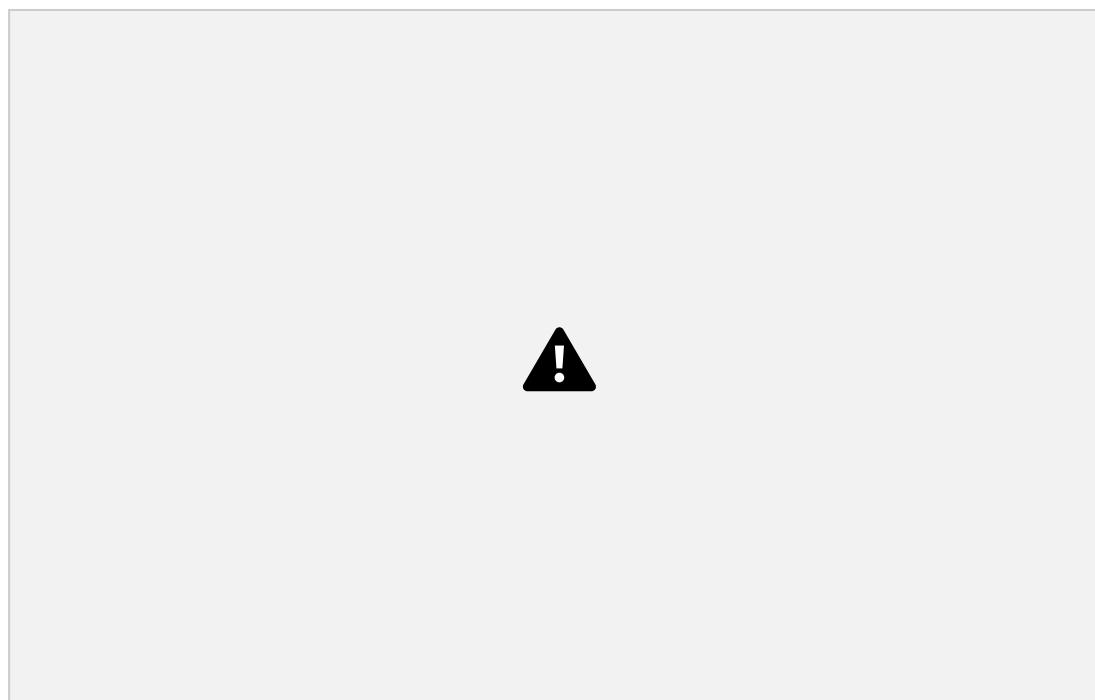
The stationarity properties of CH_SHARE were examined using the Augmented Dickey Fuller

test. The results are reported in **Table 1 of Appendix.**

The ADF test statistic of -1.7125 with a corresponding p -value of 0.4227 exceeds the critical values at 10%,5% and 1% significance levels. The critical values are -3.478 (1%), -2.882 (5%) and -2.578 (10%). Since the test statistic is higher than all critical values, we fail to reject the null hypothesis.

Thus, CH_SHARE is non-stationary in levels. This is consistent with the historical evolution of China's rare earth dominance. As shown in **Figure 1**, China's share grew steadily from the early 1990s onward, reflecting structural changes in the global REE supply chain and China's strategic expansion of production and processing capacity.

Figure 1: Stationarity Test for China Share



Source: Author's calculation based on USITC data

To determine the order of integration, the first difference of CH_SHARE was tested. Results are presented in **Table 2 of Appedix.**

The ADF test statistic is -17.4357 with a p -value of 0.0000, which is far below all critical

values. Therefore, the null hypothesis of a unit root is rejected. This indicates that the first

difference of CH_SHARE is stationary. The coefficient of the lagged first difference term is -1.3895, with a highly statistically significant t- statistic of -17.4357. The constant term is not statistically significant. (p-value = 0.2661). Thus, CH_SHARE is integrated of order 1, I(1).

Upon obtaining the ADF test statistic for the first difference of CH_SHARE, we were able to reject the null hypothesis of a unit root, i.e., the series is I(1). This is consistent with the real-world persistent upward trend of growth of share of China's exports. The changes in its share from one period to another are stable and mean-reverting. In other words, while China's dominance has grown structurally.

The International Energy Agency has repeatedly stressed that China dominates the processing and refining side of critical mineral supply chains. In its 2025 critical minerals outlook, the IEA states that China is the dominant refiner for 19 of the 20 minerals analysed, with an average market share of around 70%, and that supply growth in rare earths and other strategic minerals continues to come disproportionately from China. This dominance has sustained despite the 2010 WTO dispute and losses from the COVID-19 pandemic.

The empirical results also align with US policy concerns across administrations. Under the Trump administration, the White House stated that critical minerals are indispensable to the U.S. economy and national security, and producers depend on foreign countries to supply and process them, and that the government would work to reduce reliance on mineral imports and protect supply chains from "predatory Chinese action." During the Biden administration, DOE testimony in 2024 said China maintains a dominant position in the midstream processing of REEs and this specific dependence on a single source leaves the U.S. and its allies vulnerable to supply chain disruptions.

Taken together, the results as well as official reports suggest a monopoly-like dominance in the REE sector enjoyed by China over the years. Thus, China's position is perceived as a threat by the United States and its allies, leading to funding of projects and agreements amongst the allies

to overthrow this domination.

4.1.2 AUGMENTED DICKEY FULLER TEST FOR AUS_SHARE

The results for ADF test on AUS_SHARE are reported in **Table 3**.

The ADF test statistic for AUS_SHARE is -3.2446 with a p-value of 0.0195, which is below the 5% significance level. The critical values are -3.478 (1%), -2.882 (5%) and -2.578 (10%). Thus, we reject the null hypothesis of a unit root at 5% significance level.

Thus, AUS_SHARE is stationary in levels, i.e., $I(0)$. Therefore, Australia's export share is stationary in levels, meaning they do not exhibit the same persisting trend as China's share.

Figure 2 shows that Australia's share fluctuates over time rather than following a long-term trend. These fluctuations correspond to periodic expansions and contractions in Australia's production capacity and market participation. This implies that Australia's participation in the REE market is more volatile but less structurally dominant compared to China.

Figure 2: Stationarity Test for Australia Share



Source: Author's calculation based on USITC data

The contrasting stationarity properties of CH_SHARE and AUS_SHARE reflect the different roles shaped by policy and market structure in the global REE market. China's behaviour of non-stationary in levels but stationary in first differences can be closely linked to China's long-term industrial policies, including 1999 and 2005 export quotas, and the broader state-level consolidation under the "National Rare Earth Strategy" for REE mining and processing.

On the other hand, Australia's policies have been more market driven as a response to supply vulnerabilities faced by the world and majorly the United States, since 2010. For instance, efforts to revive the Lynas Rare Earth (Australian company) supported through partnerships with the US Department of Defence and initiatives under the US Critical Minerals Strategy, aimed to diversify supply chains away from China. However, these efforts have been dependent on geopolitical shocks rather than sustained industrial policy dominance. Consequently, while China's export trajectory reflects a policy driven structural transformation in supplying major markets like the US, Australia's participation can only be understood as efforts to position itself

as a strategic “alternative” supplier in the face of geopolitical rivalry.

4.2 Regression Results

4.2.1 AR (1)

An autoregressive model of order one [AR(1)] was estimated. The results are reported in **Table 5 of Appendix**.

The coefficient of D_CH_SHARE is -0.3107 and remains statistically significant at the 1% level ($p = 0.0000$). The AR(1) term has a coefficient of 0.7629. The constant term remains significant ($p = 0.0062$). This suggests that past values of Australia's export share strongly influence current values. This is also consistent with the nature of REEs, which require gradual investment and heavy infrastructure development as opposed to other goods.

The R-squared value is 0.60, and the Durbin-Watson value is 2.33, highlighting that Australia cannot immediately respond to shifts in China's dominance even when the global demand or

geopolitical conditions, such as US efforts to diversify the supply chain, create opportunities. Instead, adjustments occur with lag, reflecting structural rigidity and capacity constraints.

Thus, the results suggest that China's expansion exerts immediate comparative pressure, and Australia's response is gradual and constrained by real economic frictions, reinforcing its role as a delayed alternative supplier rather than a second mover competitor.

4.2.2 ARDL MODEL ESTIMATION

An Autoregressive Distributed lag model was estimated with optimal lag selection based on the Akaike Information Criterion. The selected specification is ARDL (4,4), as shown in **Table 6 of Appendix**.

The model includes 4 lags of AUS_SHARE and CH_SHARE.

Among the lagged dependent variables, the first and the fourth lag of AUS_SHARE are statistically significant. ($p = 0.0000$ and $p = 0.0267$).

The CH_SHARE variable has a coefficient of -0.6484, which is statistically significant at the 1 % level ($p = 0.0000$). This indicates a strong negative short run relationship between China's export share and Australia's export share. This suggests that a one unit increase in China's share leads to a substantial contraction in Australia's share within the same period. This indicates that China's expansion, enabled by scale economies, processing dominance and pricing power, poses a threat to emerging suppliers from entering the market.

The model has an R-squared of 0.6955 and an adjusted R-squared of 0.6737, indicating a good fit. The F - statistic is 31.9723 ($p = 0.0000$), confirming overall model significance. The Durbin Watson statistic is 2.0448, indicating no autocorrelation.

The significance of AUS_SHARE (-1) and AUS_SHARE(-4) highlights that Australia's export share is not only persistent but also exhibits delayed adjustment. The supply responses from

Australia are distributed over time. When Australia's export share rises or falls, the impact tends to persist for several quarters because the industry cannot adjust quickly, mining projects take time to expand or scale down, and firms are often tied into long term supply contracts. To put it simply, Australia is unable to immediately respond to changes in market conditions because infrastructure changes require significant investment and approval.

These results imply that China exerts immediate and substantial downward pressure on Australia's export share while Australia's response is gradual and constrained by structural factors. This reinforces the interpretation that Australia is not seen as an equal competitor but as a catching-up supplier whose gains are contingent on temporary opportunities created by China rather than sustaining independent expansion. (**Figure 3**)

Figure 3: Comparative trends in China's growth and Australia share



Source: Author's Calculation based on USITC data

This intuition is important in understanding the effectiveness of the US's efforts to diversify imports away from China to allied members like Australia. Despite multiple projects to revive the industry in either country, the results from the study show that Australia has yet to prove itself as a viable option for the US in the long run. Many reasons can be attributed to Australia

lagging despite sustained efforts, but the simplest explanation for this phenomenon is the fact that both the US and Australia still operate on strict regulations and policies for the development of this industry, which leads to a lot of bureaucratic rigidity due to the nature of REE processing and their application in vital industries like defence.

In contrast, China's expansion in the 1990s was facilitated by regulatory flexibility, lenient environmental policies and targeted industrial policy, which enabled China to scale production and consolidate control over the supply chain at a pace that more tightly regulated economies like the United States and Australia are unable to match.

4.3. Bounds test

The existence of a long run relationship was tested using the ARDL Bounds Testing approach. Results are reported in **Table 7 of Appendix**.

The resulting F - statistic is 2.4604 04. The critical bounds at the 5 % level are 3.62 (I(0)) and 4.16 (I(1)). Since the F - statistic is below the lower bound, the null hypothesis of no level relationship cannot be rejected. This implies that although a short run comparative effect exists between China and Australia, there is no statistically significant long-run equilibrium relationship between their export shares.

This finding suggests that changes in China's dominance do not permanently determine Australia's market share; instead, Australia's participation over the years is influenced by other structural factors, government policy, technology shifts and global diversification efforts. Australia does not automatically gain when China loses, meaning that the United States cannot rely on a predictable substitution between China and Australia. The results suggest that the US import diversification strategy away from China is not stable in the long-run.

4.5 Model Fit

The actual versus fitted graph (**Figure 4**) shows that the fitted values track the general movement of Australia's export share relatively closely. The residuals fluctuate around 0, indicating the model captures much of the systematic variation in the data.

Some deviations occur during periods of sharp changes in export shares, reflecting external shocks such as demand surges and supply disruptions.

Figure 4: Actual vs Fitted vs Residual Values



Source: Author's Calculation based on USITC data

Chapter 5

5.1 Conclusion

This study set out to examine the impact of geopolitical tensions, particularly between the United States and China, on the trade dynamics of REEs, with a specific focus on export patterns, market adjustments, and short-run versus long-run effects. Thus, by combining quantitative

econometric analysis with qualitative geopolitical context, the research aimed to provide a comprehensive understanding of how strategic commodities behave under conditions of economic conflict.

While the United States has undertaken efforts to reduce dependence on Chinese REE supplies through domestic investment, partnerships with allied nations, and policy initiatives, China continues to retain a dominant position due to its established production capacity, processing infrastructure, and cost advantages. This asymmetry underscores the structural constraints faced by countries attempting to rapidly modify critical mineral supply chains.

Recent U.S. policy has attempted to address this through subsidies for domestic refining facilities, as well as collaboration under initiatives such as the Minerals Security Partnership. However, the scale and speed of these interventions may still be insufficient given the capital-intensive and environmentally sensitive nature of REE processing. As a result, complete decoupling appears neither feasible nor efficient in the short to medium term.

Amid ongoing geopolitical tensions, U.S. engagements and tensions in resource rich regions such as Iran and Venezuela have contributed to disruptions in global energy and commodity flows. Within this context, China has been able to consolidate its position across critical mineral supply chains, leveraging both its existing dominance and the uncertainties created by such conflicts. These developments align with the study's findings, where short-run shocks alter trade patterns without fundamentally restructuring long-term dependencies, highlighting how geopolitical disruptions can reinforce rather than displace existing concentrations of market power.

In conclusion, this study demonstrates that while geopolitical tensions significantly influence rare earth trade, their effects unfold within rigid structural constraints that limit rapid transformation. The case of REEs highlights that strategic autonomy in critical minerals is not achieved through short-term policy shocks but through sustained, long-term investment in capacity, technology, and international cooperation. What this implies is that the path for emerging economies lies not in complete decoupling, but in carefully managed diversification and gradual supply chain reconfiguration.

APPENDIX

TABLE 1

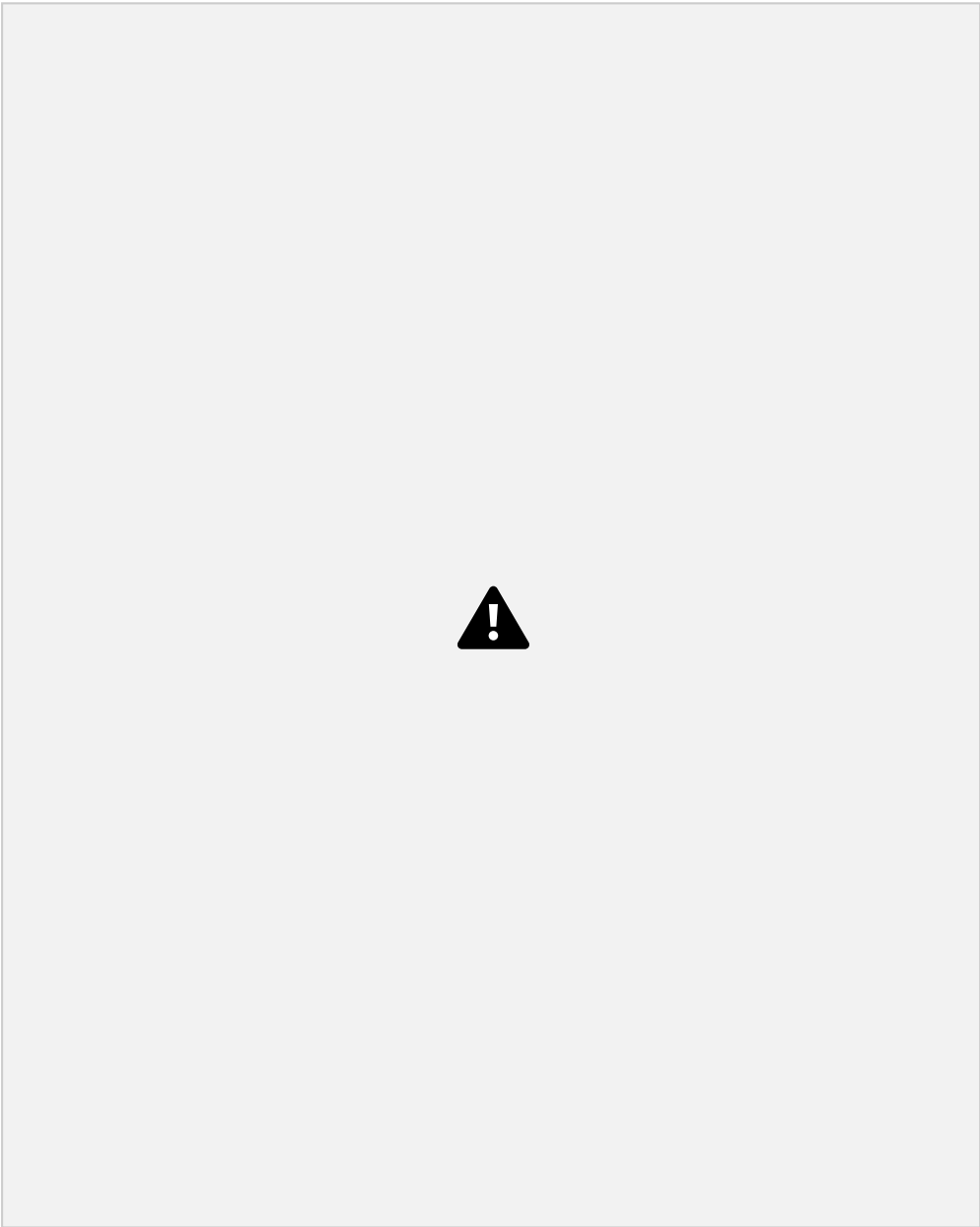


TABLE 2





TABLE 3



TABLE 4

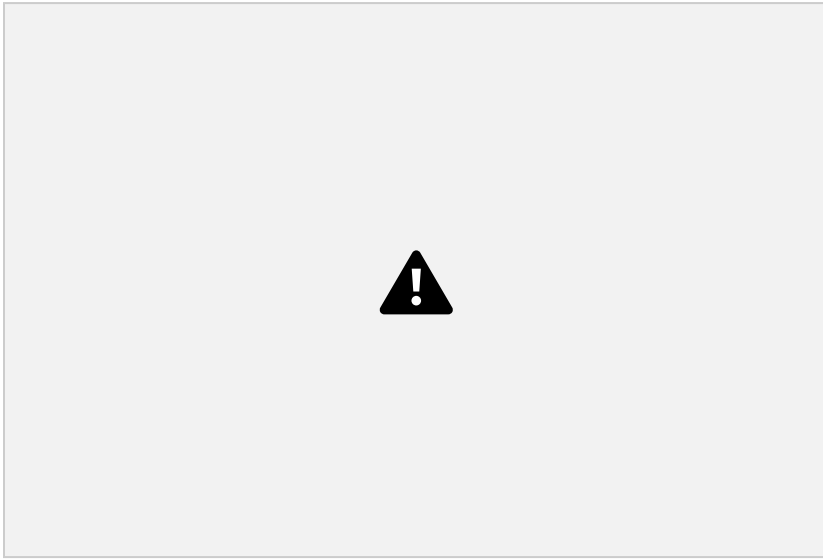


TABLE 5

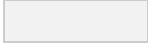
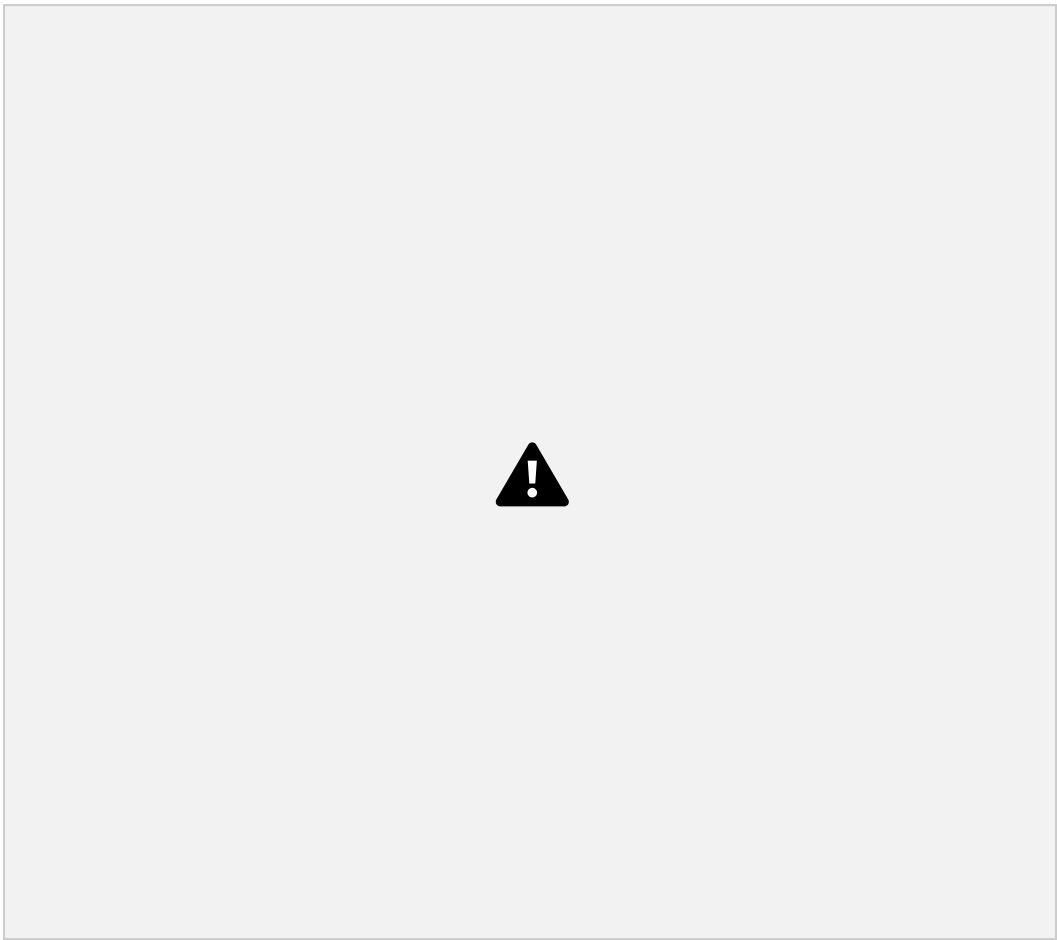


TABLE 6



TABLE 7

